APMOPS 2015 Round 1

Time Duration: 2 hours

1. In a 200-metre circular track, Alex and Ben started at the same position simultaneously. Both of them run in clockwise direction, it is given that Alex runs 6m/s whereas Ben runs 4m/s. Hence, within 16 minutes, how many times does Alex overtake Ben?

2. The ratio of the length, width and height of a cuboid is $4:3:2$, the total length of all its sides is 72cm, hence what is the volume of the cuboid?

3. Calculate:

$$(101+234+567)\times(234+567+89)-(101+234+567+89)\times(234+567).$$

4. There are 10 red, 10 yellow and 10 blue balls in a bag, these balls are identical in size. The red ball is labelled "4", yellow ball is labelled "5" and the blue ball is labelled "6", Alex takes 8 balls from the bag and the sum of the numbers on the balls is 39. What is the maximum possible number of red balls which is taken by Alex?

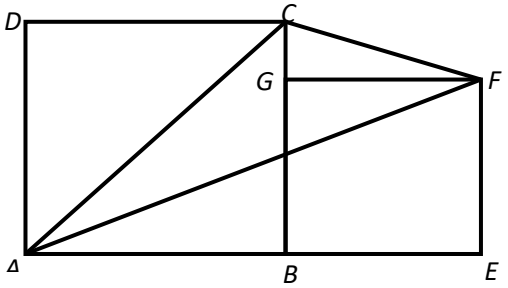
5. It is given that A is a positive integer and satisfies the following condition: $\frac{5}{9} < \frac{9}{A} < 1$. How many different values could A be?

6. It is given that there are 31 days in a particular month and the number of Monday is the same as the number of Friday hence what is the day of 10th of this month?

7. It is given that the last four digits of the product of a positive integer and 411 is 2015, hence what is the smallest possible value of the positive integer?

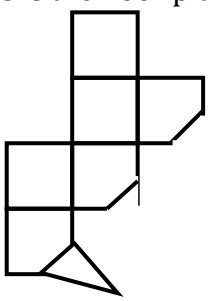
8. As shown in the diagram, a football is made up of 32 pieces, every white piece is a regular hexagon and every black piece is a black pentagon. Every white piece is adjacent to three black pieces and three white pieces; every black piece is adjacent to five white pieces. Hence how many white pieces are there?

9. As shown in the below diagram, the side of the square $ABCD$ is 8cm. Point E is located on the extended line of AB, rectangle $BEFG$ is a square, hence what is the area of triangle AFC?



10. There a kind of “5plus1” soda drink, every five empty bottles can be used to exchange for one new bottle of drink. In a party, a class drinks 109 bottles of the drink and some of the drink is gotten by exchanging empty bottles. Hence how any bottles did they buy initially?

11. As shown in the below diagram, this is the floor plan of a polygon. How many sides are there in this polygon?



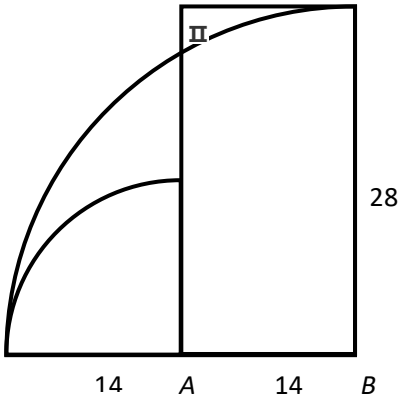
12. It is given that 2^{29} is a nine-digit number made up of nine different one-digit numbers and hence which number does not appear?

13. There are 52 students in a class, and 30 among them can swim, 35 among them can ride a bike and 42 among them can play Ping-Pong, hence at least how many students can play all three sports?

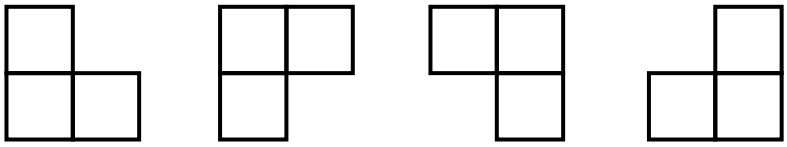
14. A company plans to transfer 89 tons of goods each large van can transfer 7 tons and each small van can transfer 4 tons. If every large van requires 14 litres of fuel and each small van requires 9 litres, at least how many litres of fuel is required to transfer all the goods?

15. Assume the sum of internal angles of convex polygon after deducting away an internal angle is 2015°, what is the degree of this particular internal angle?

16. As shown in the below diagram, A and B are the centres of the quarter circles respectively, the radius of two circles are 14cm and 28cm. please calculate the difference between area I and area II. ($\pi = \frac{22}{7}$)

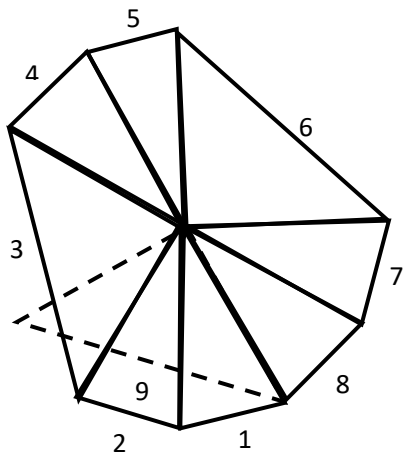


17. In a 6×6 form, how many shapes as shown in below diagram can be found?

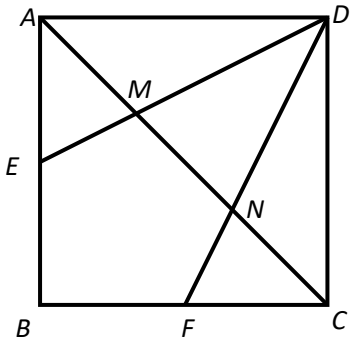


18. A natural number is divided by another natural number, the quotient is 15 and the remainder is 5, it is given that the sum of the divisor, dividend, quotient and the remainder is 2169, hence what is dividend

19. There are two isosceles triangles with side of 1, one of them is acute-angle triangle with vertex angle of 30° and the other one is a right-angle-triangle with vertex angle of 90°. As shown in the below diagram, now put these two types of triangles on a point in clockwise direction. The rule is as following: the first and second is acute-angle triangle and the third is a right-angle triangle; the fourth and fifth are acute-angle triangle and the sixth is a right-angle triangle. As such, the Nth triangle coincides with the first triangle perfectly. Calculate the minimum value of *N*.



20. As show in the below diagram, *ABCD* is a square, point *E* and *F* are the midpoint of *AB* and *BC*.*DE* and *DF* crosses *AC* at *M* and *N*. If the area of the square *ABCD* is 48cm^2, please calculate the area of *EBFNM*.

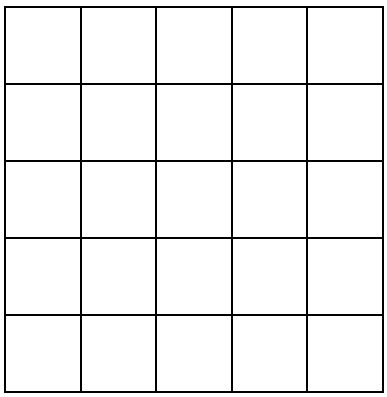


21. There are four football teams. There is a race between every two teams. A team gets three marks for winning a game, gets 0 mark for losing a game and gets 1 mark for a tie. It is given that the marks of four teams after completing all the races are exactly four consecutive positive integers, what is the product of these four numbers?

22. There are five water tapes attached to a water pool. To fill the pool, the first four tapes need to be opened for six hours; or the last four tapes need to open together for 8 hours; if only the first and the last tape open, they need 12 hours to fulfil the pool. Hence how many hours it needs for last tape alone to fulfil the pool.

23. How many consecutive "0" are there at the end of the product of $5 \times 10 \times 15 \times \cdots \times 2010 \times 2015$.

24. In the below diagram, paint some small squares in a way that within every 3×3 large squares, there are exactly four painted squares, hence at least paint how many squares?

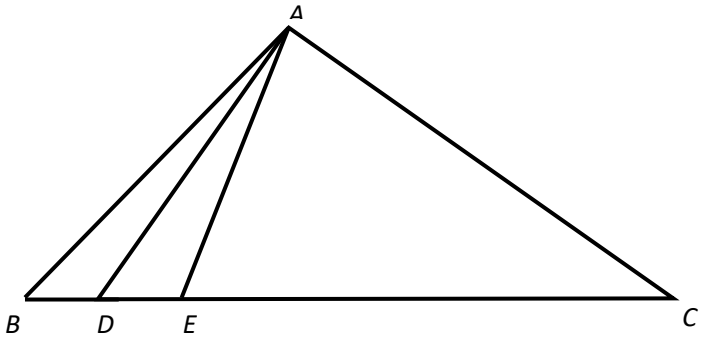


25. From balls that have label of 1 to 49, pick some balls to form a circle in a way that the product of numbers on the adjacent balls is less than 100, hence at most how many balls can we pick?

26. Car A and B are moving from Station A and B in the opposite directions, the ratio of speed of Car A and B is3:2. Station C is located between Station A and B, the time for Car A and B reaches C are 9:00 and 19:00 respectively. What is the time Car A meets Car B?

27. Assume a ,b and c are numbers from 0 to 9 (allows to be the same), if to simplify all numbers in form of $0.\dot{a}b\dot{c}$ into fractions, how many different values the numerator can be ?

28. Put cards A, K, Q, J and 10 into five envelops labelled A, K, Q, J, 10. Each envelop contains a card, how many ways are there for the every card to be put into the wrong envelop?

29. How may 5-digit number can be divisible by 3 and contain at least a number "3"?

30. As shown in the below diagram B, D, E and C are on the same line. It is given that $\angle BAD = \angle DAE = 12°$, AC and AD are perpendicular to each other, $BC = AB + AE$, what is the degree of $\angle ABC$?